

Matthew Topham

AI Systems Architect & Full-Stack Software Engineer

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I build interactive AI products, persistent world systems, and multimodal pipelines that turn AI roleplay into reference-grounded animated scenes. Active Marinara Engine developer with 100+ merged pull requests across React/TypeScript, Node.js, capability APIs, transactional state, AI generation, responsive UI, and production validation.

Open to opportunities: AI engineering, full-stack software, interactive media, and flexible roles compatible with completing my computer science degree.

TECHNICAL SKILLS

TypeScript JavaScript Python SQL
React Node.js React Query Vite
Postgres REST APIs WebSockets
Docker GitHub Actions Playwright
Linux Unity ComfyUI

AI systems: LLM integrations, agent/runtime architecture, reference-grounded storyboard orchestration, image-aware motion planning, ComfyUI/MiniMax H3, local AI.

Reliability: transactions, idempotency, revision conflicts, race handling, snapshots/checkpoints, regression testing.

EDUCATION

University of Colorado Boulder

B.A. Computer Science - Expected May 2027

Front Range Community College

A.S. Computer Science - 2025

CERTIFICATION

NVIDIA-Certified Associate

AI Infrastructure & Operations - Aug 2025

EXPERIENCE

Pasta Devs - Marinara Engine

Remote | 2026-Present

AI Systems Architect & Open-Source Developer

Active developer on a local-first AI chat, roleplay, and game engine with downloadable agents, persistent worlds, multimodal generation, and multiple model providers.

- Authored 100+ merged pull requests across React/TypeScript UI, Node.js services, shared contracts, package APIs, transactional persistence, prompts, documentation, and automated tests.
- Shipped an end-to-end Roleplay-to-animation pipeline: completed exchanges become storyboard episodes; Character Sheets and World Maps artwork ground people and locations; generated T=0 keyframes feed image-aware motion planning; local MiniMax H3/ComfyUI renders produced a live-tested multi-shot clip with synchronized motion and audio.
- Architected hierarchical world-state infrastructure with nested locations, graph validation, bounded prompt context, AI drafting/expansion, lore grounding, snapshots/checkpoints, revision conflict handling, and responsive map authoring.
- Built reliable movement workflows using atomic owner turns, idempotent command IDs, server-owned route validation, retry/recovery semantics, and stream/persistence sanitization for hidden model directives.
- Extended the downloadable capability runtime with versioned host contracts for model routing, resources, persistence, transactions, history/checkpoints, readiness, retries, and package-owned client contributions.
- Validated changes with deterministic regression suites, Playwright browser tests, lint/type/build gates, bundle budgets, container checks, and manual desktop/mobile QA.

Selected proof:

[Roleplay animation #4840](#) | [Capability API #3693](#) | [World state #3565](#)
[Storyboards #4326](#) | [MiniMax H3 prompts \(Marinara Agents #365\)](#)

SELECTED PROJECTS

School of the Ancients - Creator & Full-Stack Developer

Voice-driven Socratic learning platform where users converse with AI historical mentors across web and VR experiences.

- Implemented React/Vite interfaces, Node.js and Supabase services, authentication, RLS-backed Postgres, lesson and quest creation, CI/CD, real-time voice AI, and Unity XR prototypes.
- Built web and VR interaction paths around conversational AI, educational quests, and persistent user progress.

Project: github.com/School-of-the-Ancients

AI Radar - Creator

Python and GitHub Actions pipeline that ingests RSS feeds, filters and summarizes AI research/news with LLMs, and publishes a concise daily brief.

Repository: github.com/thetopham/ai-radar

ADDITIONAL ENGINEERING

- Built modular trading automation with Python, real-time SignalR feeds, API-based order routing, Supabase logging, and model-assisted analysis.
- Automated Linux/Cosmos infrastructure backups, monitoring, alerting, and failover workflows.
- Developed local AI workflows spanning LLM serving, ComfyUI image/video generation, prompt pipelines, and GPU-backed inference.

SELECTED MARINARA SYSTEMS SHIPPED

- **Reference-grounded animation:** Character Sheets and World Maps references feed generated first-frame keyframes, image-aware shot planning, and local image-to-video rendering.
- **Persistent worlds:** Hierarchical locations, stable IDs, validated movement, lore grounding, reusable/shared worlds, map artwork, route planning, and checkpoint-aware state.
- **Capability runtime:** Versioned package contracts and host services for model routing, persistence, transactions, resources, history, logging, readiness, and client lifecycle.
- **Storyboard Agent:** Separate Game/Roleplay planning, provider-neutral prompt stages, editable templates, image-aware motion refinement, and local MiniMax H3 prompt paths.
- **Reliability:** Deterministic regressions, package lifecycle testing, Playwright desktop/mobile validation, stream sanitization, conflict handling, retry/recovery, and race-safe persistence.

LINKS

- GitHub:** github.com/thetopham
- Marinara Engine:** github.com/Pasta-Devs/Marinara-Engine
- Marinara Agents:** github.com/Pasta-Devs/Marinara-Agents
- Marinara contributions:** github.com/Pasta-Devs/Marinara-Engine/pulls?author=thetopham
- Portfolio:** thetopham.github.io
- LinkedIn:** www.linkedin.com/in/mrtopham